

3rd Workshop on Explainability in Human-Robot Collaboration: Real-World Concerns

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Abstract—Robots powered by AI and machine learning are increasingly capable of collaboration and social interaction with humans, leading to a demand to develop new approaches to ensure their transparency and explainable behaviour. As explainable AI (XAI) seeks to clarify AI decisions, its integration into physical robots often creates an *illusion of explainability*—raising questions about whether current approaches truly enhance understanding. The 3rd Workshop on Explainability in Human-Robot Collaboration aims to address the *real-world concerns* associated with developing explainable and transparent robots through a focused, multi-faceted panel discussion and a series of paper presentations. In this workshop, we will focus on refining when and how explanations should be provided, integrating human communication principles to enhance trust and transparency in human-robot collaboration through both technical and user-centred solutions.

Index Terms—Explainable Robotics, XAI, Human-Centered Robot Explanations

I. INTRODUCTION

Robots, increasingly powered by AI and machine learning, are becoming capable of collaborative actions and social interactions with humans. However, the complexity of these systems demands new approaches to ensure transparency and comprehensibility. The field of Explainable AI (XAI) has emerged to address this need, aiming to make AI decisions more intelligible [1]. Yet, when AI systems are embedded in physical robots that interact with humans, explainability becomes more challenging [2]. Humans often attribute human-like traits to robots, leading to expectations of personalized, interactive explanations, particularly when anomalies occur in the interaction [3], [4]. In this context, providing explanations is not just a system requirement but a communicative act within the interaction.

Many current approaches treat explanations as static, statistical outputs detached from the dynamic environment and user needs. However, robotic transparency should extend beyond utility to address ethical concerns, such as privacy, and empower users with their “right to an explanation” [5]. The 3rd version of the explainability in HRI workshop aims to further

explore real-world concerns of interactive robot explanations, focusing on when and how explanations should be provided and how they can incorporate social characteristics inspired by human communication principles.

To achieve this, we recognize that providing explanations constitutes a social act typically involving multi-modal communication [6], [7]. Therefore, a multidisciplinary approach is fundamental. To bridge the gap between technical solutions and effective human-centred solutions, we aim to bring together not only experts in explainable AI, engineering and computer science but also experts in human cognition and social sciences [8], [9].

II. WORKSHOP OVERVIEW

The goal of this workshop is to deepen the conversation on explainability in Human-Robot Interaction (HRI) by bringing together insights from both humanistic/social and technical perspectives. We aim to challenge the assumptions surrounding AI transparency in HRI, particularly focusing on the concept of explainability. We aim to explore these challenging questions through a comprehensive panel discussion with scholars from different areas of expertise and a series of lightning talks. Our panel, titled “*The Illusion of Explainability—Are We Misleading Ourselves About AI Transparency in Real-World HRI?*”, brings together four distinguished experts in the field, each delivering a 10-minute talk addressing key issues before opening the floor to an in-depth panel discussion.

The objective of the workshop is to systematically explore the following topics:

- Whether current AI transparency and explainability in HRI genuinely promote understanding of robot behavior or merely create an illusion of explainability.
- The gap between theoretical explainability frameworks and their application in real-world HRI scenarios, where the complexity of interactions often overshadows clear communication.

- The role of embodiment and context in shaping what is perceived as explainable behaviour in robots and how this perception influences trust and collaboration.
- What are our responsibilities as researchers and developers while developing explainable and transparent AI systems?
- Whom to blame, for the unintended side effects? The robot, the human, the lack of explanations, or the illusion of explanations?

Through the diverse perspectives of our panellists and subsequent discussions, we aim to critically assess the state of explainability in HRI and identify future directions for creating genuinely transparent and trustworthy AI systems in practice.

A. Main Topics of Interest

Topics of interest include, but are not limited to:

- Using participatory design to achieve explainability
- The downsides of explainability
- The connection between explainability and trust
- What makes an interaction explainable
- Metrics to evaluate explainability
- Deception in Human-Robot Interaction
- Unintended biases in explainability & how to alleviate them
- Transparency in Trustworthy autonomous systems
- Explanation generation as a model reconciliation process
- Adapting explanation through forming a mental model
- Explanation generation
- AI Ethics
- Bias and misinformation in LLM-based explanations
- Few-shot learning and adaptation for LLM in explainable HRI

B. Target Audience and Solicitation Plan

We aim to engage a diverse group of individuals interested in explainable human-robot collaboration. We encourage participation from researchers, robot users, and stakeholders across fields such as human-robot interaction, robotics, psychology, artificial intelligence, and other interdisciplinary areas, particularly those investigating either the technical or philosophical dimensions of explainability and transparency in HRC. Industry professionals involved in the design and development of collaborative robots are also welcome to join the workshop. This is the third version of the workshop, following two successful editions conducted at RO-MAN 2023 and HRI 2024, each with 30-40 participants. Given that the workshop at HRI 2024 reached full capacity, we anticipate the number of participants would be around 40-50 for this edition.

The call for participation will be distributed via mailing lists and social media channels, including but not limited to HRI-Announcement, CHI-Announcements, and Robotics Worldwide. In addition, we extend the invitation to mailing lists outside the direct scope of human-robot interaction, such as Corpora-list (aimed at NLP researchers), CVML (aimed at researchers working on computer vision and machine learning), HMC (aimed at researchers working on human-machine communication), the Women in Machine Learning mailing list, and the Women in Robotics Slack channel.

A website modelled on the previous [10]¹ workshops will be created to provide detailed information. The website will provide the workshop format, schedules, and panel information. Participants will be invited (but not required) to submit 3 to 4 pages (including additional pages for references and appendices) of extended abstracts related to the topics described above. Submissions of work in progress and preliminary results are also encouraged. All submissions will be peer-reviewed based on criteria such as originality, relevance, technical soundness, and clarity. Accepted submissions will be invited to give a 10-minute presentation, including the Q&A session with a emphasis on providing a platform to discuss the submissions.

C. Invited Panellists

Matthew Gombolay (Confirmed), Associate Professor of Interactive Computing at the Georgia Institute of Technology (USA). His research focuses on democratizing robotics by designing robots that can learn from interactions with people and their environments, communicate the result of their learning to human partners, and collaborate effectively in diverse human-robot teams.

Dana Kulic (Confirmed), Professor at the Department of Electrical and Computer Systems Engineering and the Department of Mechanical and Aerospace Engineering of Monash University (Australia). Her research focuses on autonomous systems that can collaborate with their users using natural and intuitive interaction strategies that are learned directly from the interaction with their users.

D. Format and Activities

This half-day workshop will be conducted in a hybrid format, offering both in-person attendance and Zoom access to ensure broader accessibility. To support remote engagement, a designated member of the organizing committee will actively monitor the chat and encourage contributions from online participants. Authors of extended abstracts will present their research during the lightning talk session at the beginning of the workshop. These short, focused presentations will allow participants to briefly introduce their research and insights to the group, fostering a foundation for discussions throughout the day. After the coffee break, the workshop will continue with presentations from the panelists, who will explore key themes related to the workshop's central topic, *'The Illusion of Explainability — Are We Misleading Ourselves About AI Transparency in Real-World HRI?'*. This will be followed by a panel discussion, bringing together diverse viewpoints from both technical and social fields, fostering a deeper exploration of current approaches, challenges, dilemmas, and key questions surrounding the use of explainability in HRI. During this session, participants will have the opportunity to interact with the panelists, ask questions, and contribute to the conversation. The event will close with concluding remarks, summarizing key insights from the talks and discussions, leaving attendees with potential directions for future research and collaboration.

¹<https://sites.google.com/view/x-hri>

E. Structure

- 09:00 - 09:05** Welcome and introduction to the workshop
09:05 - 10:30 Lightning talks (8 * 6 min + 4 min)
10:30 - 11:00 Coffee Break
11:00 - 11:40 Panellists Talks (40 min)
11:40 - 12:40 Panel (60)
12:40 - 13:00 Concluding remarks

III. DOCUMENTATION

The workshop's outcomes will be documented on our website, <https://sites.google.com/view/x-hri/home> and will include submitted papers, abstracts, and event logs generated from the discussions. Furthermore, we hope to organize a special issue on the topics related to explainability and transparency in human-robot collaboration in a related journal, given the success of this workshop series and the research surrounding this topic. All accepted authors and other participants will be invited to contribute.

IV. ORGANIZING COMMITTEE

Elmira Yadollahi is an Assistant Professor at Lancaster University, UK. Her research tackles explainability in robotics and multimodal perception and multimodal explanation to enhance the transparency and efficiency of the collaboration between humans and machines. She has organized the 2nd Workshop on Explainability at HRI 2024, as well as a series of workshops on Child-Robot Interaction at ICSR 2020, IDC 2021, IDC 2024, and another series on Interactive AI at ROMAN 2024. She is an associate editor of IJCCI Journal and has served as chair and AC in IDC, ICSR, AAMAS, and HRI.

Fethiye Irmak Dogan is a Postdoctoral Research Associate at the University of Cambridge, UK. She received her Ph.D. degree in Computer Science from KTH Royal Institute of Technology, Sweden, in 2023. During her doctoral studies, she was also a visiting scholar at the Georgia Institute of Technology, US (Nov 2021 – Apr 2022). She holds M.Sc. (2018) and B.Sc. (2015) degrees in Computer Engineering from Middle East Technical University, Turkey. Her research aims to deploy autonomous robots in human environments to facilitate people's lives, and her areas of interest include human-robot interaction, machine learning, and deep learning.

Marta Romeo is an Assistant Professor in Computer Science at Heriot-Watt University, Edinburgh (UK). She received her PhD in socially assistive robotics for independent living within the H2020 MoveCare project at the Cognitive Robotics Lab at the University of Manchester (UK). Her current research, within the UKRI TAS Node on Trust, focuses on investigating how trust between humans and robots can be built, maintained, and recovered. She is interested in transparency and communicating robot's intentions in a multimodal way.

Dimosthenis Kontogiorgos is a Postdoctoral Researcher at the Massachusetts Institute of Technology, and his research explores how to develop explainable multimodal AI technologies for collaborative interactive robots. He has co-organised the Robo-Identity series of workshops at HRI 2021-2024, as well as workshops for Interspeech 2017, SigDial 2019, ICMI 2020,

and CUI 2023. He has also been involved in the organising committee of ICMI 2024 and AC editorial roles for ROMAN 2024, ACII 2024, CHI 2025, RA-L, and THRI.

Peizhu "Pam" Qian is a final-year Ph.D. candidate and Future Faculty Fellow at Rice University, Houston, Texas USA. She develops explainable AI algorithms that leverage interactivity and human learning theories. Her work appears at top venues such as TPAMI, HRI, AAMAS, AAAI, and ICRA. Her work on interactively explaining robot behavior to human collaborators won best video award at AAMAS'22.

Yan Zhang is a Ph.D. student from The University of Melbourne, supervised by Dr.Wafa Johal, Dr.Jorge Goncalves, and Dr.Jarrold Knibbe. Previously, she was a Research Engineer at the Institute for AI Industry Research, Tsinghua University. She received a bachelor's degree in Systems Engineering from The City University of Hong Kong. Her current research focuses on natural language communication in real-world Human-Robot Collaboration. She has publications in CHI, ICRA, and IROS.

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